Band-1 Hadamard Order 6436 — Miyamoto (Paley-I H_1608 + C_1610 over prime 1609); IN SPK via Paley-II (q=3217); Gap 2052 Still OPEN

H_1608/C_1610/H_6436 Miyamoto PASS; H_6436 Paley-II PASS (classical SPK); Paley-I/Kronecker SKIP; 6436 IN SPK via Paley-II (NOT Miyamoto-only); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_6404_Kronecker`; `2026-09-15_Band1_Hadamard_6388_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_6436_miyamoto/`.
Cutoff. 2026-09-16T00:33:39Z (~8:33PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1608 and 6436 only** (verified max $|HH^\top - nI| = 0$), via Miyamoto with preferred Paley-I seed H_{1608} = Paley-I$(q = 1607)$ and symmetric Paley conference C_{1610} over prime $q = 1609 \equiv 1 \pmod 4$, **and** via classical Paley-II H_{6436} = Paley-II$(q = 3217)$. Paley-II for the seed ($q = 803 = 11 \cdot 73$) **FAIL**s; Double half=804 **FAIL**s. Classical SPK for order 6436: Paley-I ($q = 6435 = 3^2 \cdot 5 \cdot 11 \cdot 13$ not prime) **FAIL**; Paley-II ($q = 3217$ prime $\equiv 1 \pmod 4$) **PASS**; Kronecker (4×1609) **FAIL**. Therefore order **6436 is IN SPK via classical Paley-II (NOT Miyamoto-only)**; Miyamoto is an additional finite construction. Seed **1608** enters SPK via Paley-I. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. `clay_closed = false`.

Abstract

Finite constructions of orders **1608** and **6436**.

- H_{1608} = Paley-I ($q = 1607$) — preferred; Paley-II / Double SKIP
- C_{1610} = Paley conference over prime 1609
- H_{6436} = Miyamoto(H_{1608}, C_{1610})
- H_{6436} = Paley-II ($q = 3217$) — **classical SPK path**
- Paley-I for 6436: **SKIP**; Kronecker: **FAIL**

SPK: 6436 IN SPK via Paley-II (NOT Miyamoto-only). Seed 1608 enters SPK via Paley-I. Prior SPK seeds through 6404 ship (incl. 1600/1596) retained; **6388/6404 NOT folded**.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_6436.py
```

Results. overall_pass=true; units 78/78; wall 434.6137s; all max|HH^T-nI|=0.

Cutoff: 2026-09-16T00:33:39Z (~8:33PM ET)

Arithmetic

- Target $n = 6436 = 4 \cdot 1609$; 1609 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1608} and C_{1610} (Paley conference over prime 1609).
- Seed preferred: Paley-I $q = 1607$ (prime $\equiv 3 \pmod 4$); $q + 1 = 1608$.
- Paley-II seed $q = 803$ **FAIL** ($11 \cdot 73$); Double half=804 **FAIL**.
- SPK: Paley-I $q = 6435$ **FAIL**; Paley-II $q = 3217$ **PASS** → **IN SPK**; Kronecker **FAIL**.

Constructions (verified)

Object	Method	max|HH^T-nI|	Wall (s)
H_1608 Paley-I	Paley-I q=1607 (preferred)	0	3.2255
H_1608 Paley-II	SKIP q=803=11·73 not prime	—	—
H_1608 Double	SKIP q=803 not prime	—	—
C_1610	Paley conf q=1609	0	3.0069
H_6436 Miyamoto	Miyamoto(H_1608, C_1610)	0	208.4478
H_6436 Paley-I	SKIP q=6435=3²·5·11·13 not prime	—	—
H_6436 Paley-II	Paley-II q=3217 (classical SPK)	0	213.6143

SPK closure note

Seeds include prior SPK ships through **6404** lineage (**1600** Kronecker from that ship; **1596** from 6388 ship; **NOT** 6388/6404) + verified **1608** (Paley-I) + **6436** via classical Paley-II; **exclude** Miyamoto-only bonuses including **6404**/**6388**/**6196**/**5956**/**5812**/**5476**. Known count 1314. Next constructive gap **2052** (OPEN).

Quantity	Value
6436 in SPK	**True** (via Paley-II q=3217)
1608 in SPK	**True** (Paley-I)
1600 in SPK	**True** (prior Kronecker)
6404 in SPK	**False** (prior Miyamoto-only)
6388 in SPK	**False** (prior Miyamoto-only)
Next constructive gap	**2052** (OPEN)

Next Miyamoto targets (after q=1609)

q	n=4q	need H	build notes
1613	6452	1612	[]
1621	6484	1620	[‘PaleyI(q=1619)’, ‘PaleyII(q=809)’]
1637	6548	1636	[]
1657	6628	1656	[‘Double(PaleyI half=828)’]
1669	6676	1668	[‘PaleyI(q=1667)’]
1681	6724	1680	[‘Double(PaleyI half=840)’, ‘Kronecker(4x420)’]

Honesty

clay_closed = false · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no RH/P!=NP/YM/Frankl/NS/AC/Clay-closed claims · **6436 ∈ SPK via Paley-II (not via Miyamoto alone)**.